ModelIt!

Student Activity Pack

Can We Cure Cancer?

Modeling Cell Cycle Regulation, Oncogene Activation, and Immunotherapy Strategies

Grade 12L3 | NGSS HS-LS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about what causes cancer at the cellular level — what goes wrong inside a cell that leads to uncontrolled growth?**

2. **Why do you think cancer becomes more common as people age? What does age have to do with cellular mutations?**

3. **What is the difference between chemotherapy and immunotherapy in how they fight cancer?**

4. **If the immune system is designed to kill abnormal cells, why doesn't it always prevent cancer from developing?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Mutation Rate
- Carcinogen Exposure
- Inherited Genetic Risk
- Oncogene Activation Level
- Tumor Suppressor Function
- Cell Cycle Checkpoint Integrity
- Immune Surveillance Effectiveness
- Tumor Microenvironment Immunosuppression
- Angiogenesis Rate
- Treatment Response Score

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Multi-Hit Cancer Development

Sequential: Oncogene activation -> Tumor suppressor loss -> Checkpoint failure -> Immune escape

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Immune Surveillance vs. Immune Escape

Strong immune system | Tumor gradually increasing PD-L1 expression and microenvironment immunosuppression

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Checkpoint Inhibitor Therapy

Tumor with high PD-L1 | Strong but suppressed immune system | Apply anti-PD-1 checkpoint inhibitor

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Metastasis Potential — The likelihood that cancer cells will detach from the primary tumor, enter the bloodstream or lymphatic system, and establish secondary tumors in distant organs — metastasis is responsible for approximately 90% of cancer deaths and requires cells to overcome multiple physical and immunological barriers

Epigenetic Dysregulation — Changes in gene expression caused by modifications to DNA packaging (histones) and chemical marks (methylation) rather than changes to the DNA sequence itself — epigenetic changes can silence tumor suppressors or activate oncogenes without mutation, and are increasingly recognized as a parallel pathway to cancer development

Drug Resistance Evolution — The process by which cancer cell populations evolve resistance to treatment through natural selection — resistant subclones that pre-exist within the tumor or emerge through new mutations survive treatment and repopulate the tumor, analogous to antibiotic resistance in bacteria

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Every day, your body produces billions of new cells through mitosis. Dozens of those cells acquire mutations that could become cancerous, yet your immune system destroys almost all of them before they can grow. What happens when this defense system fails, and can we engineer the immune system to become an unstoppable cancer-killing machine? Use evidence from your simulation to support your answers.

1. **How does your model illustrate why cancer requires multiple mutations rather than just one, and what does this reveal about the body's defense architecture?**

2. **What does the immune escape scenario reveal about the arms race between tumor evolution and immune surveillance?**

3. **Why is the tumor microenvironment such a critical variable, and how does it change the effectiveness of treatments that would work against isolated cancer cells?**

4. **How does the concept of evolutionary resistance apply to cancer treatment, and what does your model suggest about the advantage of combination therapies?**

5. **What are the limitations of modeling cancer with ten components when each patient's tumor has a unique genomic and immunological profile?**

8 STEM Challenge: Design a Personalized Cancer Treatment Protocol

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YOUR ENGINEERING MISSION

Design a personalized treatment protocol for a patient with a specific cancer profile, integrating surgery, chemotherapy, immunotherapy, and targeted therapy based on the tumor's genomic and immunological characteristics, using model evidence to justify each treatment decision.

THE SCENARIO

A 45-year-old patient has been diagnosed with stage III non-small cell lung cancer. Genomic sequencing reveals: KRAS oncogene mutation, TP53 tumor suppressor loss, high PD-L1 expression (70%), moderate tumor mutational burden, and an immunosuppressive tumor microenvironment with high regulatory T cell infiltration. Your oncology team must design a comprehensive treatment plan specifying the sequence and combination of therapies, predicted response based on the tumor profile, and monitoring strategy for treatment resistance.

Engineering Design Thinking

1. WHAT IS THE SEQUENCE OF TREATMENTS IN YOUR PROTOCOL AND WHAT IS THE SCIENTIFIC RATIONALE FOR THE ORDER?

2. HOW DOES YOUR PROTOCOL ADDRESS THE IMMUNOSUPPRESSIVE TUMOR MICROENVIRONMENT TO MAXIMIZE IMMUNOTHERAPY EFFECTIVENESS?

3. WHAT BIOMARKERS WOULD YOU MONITOR DURING TREATMENT TO DETECT EARLY SIGNS OF RESISTANCE OR RESPONSE?

4. HOW WOULD YOU MODIFY THE PROTOCOL IF THE FIRST-LINE TREATMENT FAILS TO PRODUCE A RESPONSE WITHIN 12 WEEKS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

Explain your design using scientific evidence from the simulation:

Real Career Connection

Oncologists who diagnose and treat cancer patients earn $250,000-$500,000/year. Cancer researchers who develop new therapies at academic institutions and pharmaceutical companies earn $80,000-$180,000/year. Genetic counselors who help patients understand their cancer risk and treatment options earn $80,000-$120,000/year. Immunotherapy scientists who develop checkpoint inhibitors and CAR-T cell therapies earn $100,000-$200,000/year.